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Fractal clusters and urban scaling shape spatial inequality in U.S. patenting

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Innovation is distributed unevenly across space, yet the mechanisms behind persistent patenting inequality are poorly understood. Analyzing over a century of georeferenced U.S. patent data across eight major technological domains, we identify three consistent spatial patterns: (1) Zipfian scaling in patent output, (2) fractal clustering of innovation hubs, and (3) a 20-km scale of maximal spatial disparity, which we term the “inequality horizon.” These general patterns suggests that the geography of innovation emerges not only from urbanization or policy but from underlying generative principles. A minimal spatial model combining local reinforcement and long-range diffusion replicates all observed patterns without sector-specific assumptions. Analytical results show that the inequality horizon arises at the spatial scale where cluster growth balances internal heterogeneity. Our multiscale analysis suggest that emergent inequality in U.S. patenting stems by the interaction of urban scaling (superlinear productivity linked to city size) and fractal clustering (hierarchical spatial organization beyond administrative boundaries), pointing to general mechanisms governing the spatial dynamics of innovation.

Innovation displays hallmark features of complex systems: it emerges from decentralized interactions, evolves through feedback mechanisms, and gives rise to large-scale patterns that transcend individual technologies or policy contexts. Traditional models, however, often rely on equilibrium assumptions and focus on isolated factors such as urban concentration, knowledge spillovers, or regional specialization-overlooking how their interactions shape innovation dynamics^{1,2}. Classical spatial theories like Central Place Theory^{3,4} describe idealized, symmetric arrangements of activity, while stochastic models such as Gabaix’s explain size distributions without considering spatial structure⁵. Because these models capture different aspects of the innovation process, combining them may offer a more comprehensive explanation for why some places innovate more than others. Capturing the persistent spatial inequality of innovation requires models that move beyond scale-free assumptions and incorporate its uneven distribution across regions.

A complex systems perspective shifts the focus from system-specific factors to general mechanisms that govern innovation across sectors and regions. Prior research has shown that innovation scales superlinearly with city size-larger cities produce disproportionately more patents-inspiring the development of urban scaling theory⁶. These models typically address population-output relationships within urban boundaries. They do not, however, explain how innovation is distributed across space. As a result, the relationship between urban scaling and the spatial heterogeneity of

innovation-the emergence of dense hubs alongside sustained peripheral activity-remains poorly understood.

Filling this gap requires an approach that identifies spatial regularities across technological domains and links them to underlying generative mechanisms. From this perspective, we want to examine how innovation patterns emerge from the interaction of agglomeration, spatial diffusion, and cumulative advantage. Such analysis is now possible due to recent efforts that have produced high-quality data on the historical dynamics of patenting, including precisely geolocated information on individual inventors^{7,8}.

In this study, we analyze over a century of U.S. patent data across eight high-impact technology sectors and identify three robust statistical patterns: (1) a Zipfian distribution of patent output across locations, (2) fractal spatial clustering of innovation locations, and (3) a critical 20-km inequality horizon, the spatial scale at which patenting activity inequality peaks before decreasing. While urban scaling theory captures important aggregate trends^{5,9}, it does not address how these empirical patterns manifest across spatial scales-or how patenting inequality emerges and stabilizes. We hypothesize that patenting inequality is shaped not only by urban concentration^{10,11} but also by diffusion processes that extend beyond city boundaries.

To develop our hypothesis, we propose a minimal stochastic model in which innovation is driven by the interplay of two generative processes: local

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reinforcement (preferential attachment) and long-range dispersal. Although more elaborate models can be defined, our framework demonstrates that a simple model can account for the statistical features of innovation landscapes—without sector-specific tuning. More broadly, by framing innovation as a complex system, this model seeks to investigate the intersection of economic geography, network science, and complexity theory—specifically, how large-scale technological change emerges as the result of decentralized dynamics rather than being centrally directed. Modeling the spatial evolution of innovation networks could help policymakers develop models for identifying emerging patenting areas and designing interventions that align with the underlying structure of innovation systems.

Results

Innovation activity in the U.S. displays statistical regularities pointing to an underlying common structure. By analyzing over a century of georeferenced patent data (Methods), we find three interconnected patterns linked to the geography of innovation: a Zipfian scaling of patenting activity, a hierarchical clustering of innovation networks, and the emergence of a spatial scale at which inequality is maximized. These general patterns, when combined, have a direct influence on a simple generative model that explains them, regardless of the specifics of urbanization, policy regimes, or sector composition.

Zipf scaling of innovation output

The first signature of complexity in U.S. innovation systems is the heavy-tailed distribution of patenting activity (2), which remains stable across time, geographic regions, and technological domains (Fig. 1). This heavy-tailed distribution is indicative of cumulative advantage and preferential attachment^{12,13}. Ranking locations by total patent output reveals a Zipfian distribution with an exponent close to -1 (see Methods). This indicates that a few regions dominate innovation while many others contribute at lower but persistent rates. This skewed distribution is a common feature of complex systems, including city sizes^{14,15}, scientific citations^{16,17}, and human mobility^{18,19}.

The stability of the Zipf exponent across technological domains suggests that innovation capacity is not solely driven by urban population size but reflects more general multiplicative growth dynamics^{20–25}. Such processes generate power laws when the probability of growth is proportional to an entity's existing size. A Zipf exponent near 1 ($\gamma \approx 1$) corresponds to a slow-inequality regime, in which dominant hubs expand while smaller centers persist over time and continue to contribute²⁶. This suggests both heterogeneity and resilience.

We confirm the cumulative advantage effect in U.S. patents by analyzing the probability of patenting conditional on prior output (see Methods). Patent production rates grow proportionally to past activity, indicating a reinforcement process (see Fig. 2a). This pattern holds consistently across time, as shown by the segment-wise trends in the data, suggesting that the probability of future patenting increases steadily with the cumulative number of patents. Unlike systems that slow down as they use more resources^{6,27}, U.S. patenting displays sustained growth: each new patent raises the chance of further innovation by approximately the same amount (0.045% on average), regardless of technological domain.

To better understand how patent activity evolves, we examined the rate at which new locations enter the innovation system over time. This analysis reveals an initial boom characterized by rapid geographical expansion and high entry of new locations, followed by a gradual slowdown in which entry rates decline as incumbent hubs consolidate their advantage (see Fig. 2b, c; Methods). These phases are empirically defined by their innovation rates (rates of new entrants). Together, they reflect strong cumulative advantage as well as persistent (but diminishing) diversity of the innovation landscape.

Fractal geometry of U.S. innovation networks

While traditional studies emphasize metropolitan hubs, our findings reveal that innovation is neither “scale-free” nor confined to a single spatial scale. Instead, innovation clusters form a fractal-like hierarchy, with smaller

centers embedded within larger regional networks (Fig. 3a, b). The geography of innovation follows a nested, multi-scale structure rather than a simple core-periphery model²⁸.

To formalize this spatial pattern, we construct geometric graphs by connecting patenting locations within a fixed radius R , treating each location as a node and linking those within this spatial threshold (see Methods). By varying R , we observe how individual locations aggregate into clusters across spatial scales. The resulting clustering tree (Fig. 4) shows scale-free coalescence (6): clusters merge recursively without a single dominant hub, resulting in the self-similar branching of fractal networks^{29–32}.

To assess if this fractal clustering can be explained by urban size or population density alone, we compare the actual distribution to two null models defining limiting scenarios (Methods): (1) a random null model that removes any effects of population size, urban concentration, and spatial structure through uniform random sampling of populated areas; and (2) a population-weighted null model in which locations are chosen in proportion to their actual populations but lack innovation-specific mechanisms. This model retains the statistical effect of urban agglomeration, as described by scaling models⁶, but lacks any innovation-specific spatial coherence or path-dependent structure.

Figure 3c–e shows that neither null model reproduces the scaling exponent in the fraction of connected components $f(R)$ with aggregation radii R (4). While the second model incorporates demographic realism, it remains a static snapshot—there is no historical accumulation, no iterative growth, and no path dependence. Although agglomeration explains broad correlations between city size and innovation output⁶, it fails to reproduce the finer-scale, fractal clustering observed in real innovation data. Innovation landscapes reflect cumulative, history-dependent processes—including feedback, local reinforcement, and long-range diffusion—that are absent from equilibrium-based or cross-sectional models.

The difference between actual and synthetic clustering emphasizes the role of generative, temporal mechanisms in producing spatial innovation. These findings go beyond basic population scaling to imply that the geometry of innovation is influenced by more than just demographic concentration.

Multiscale analysis of patenting inequality

Innovation landscapes display a consistent spatial feature: a regional scale—approximately 20 km—at which spatial inequality in patenting activity peaks before declining at larger scales (see “Methods”). To identify this scale dependence, we apply a variable-radius aggregation method and compute the coefficient of variation (CV) in patent output as a function of aggregation radius R (8). At small aggregation radii ($R \leq R_{\min} \approx 3.3$ km), CV remains approximately constant. This plateau suggests that sub-3 km urban units—possibly reflecting the extent of walkable historical cores—function as internally cohesive zones with limited disparity. This observation aligns with urban design theory, which identifies neighborhoods or compact districts of 3–4 km in diameter as stable, modular units of activity³³. As R increases beyond this core scale, inequality rises, reaching a maximum around $R \approx 20$ km, and then declines as heterogeneity is progressively averaged out (Fig. 5, top row).

To better understand this transition, we examine how the rank-patent distribution evolves with spatial scale (Fig. 5, bottom row). At $R = R_{\min}$, the distribution follows a two-regime Zipf law, with distinct exponents— γ_1 for high-output locations and γ_2 for lower-output ones—reflecting moderate inequality among urban cores and peripheral areas. As R approaches the critical scale ($R \approx 20$ km), these regimes converge into a single power law, indicating that inequality becomes scale-invariant at this spatial threshold. Beyond it, a steeper Zipf exponent (γ_3) emerges, consistent with amplified disparities across merged urban regions and their surroundings.

This marks a structural shift in how innovation clusters manifest across space. Around 20 km, nearby hubs begin to merge into broader regional

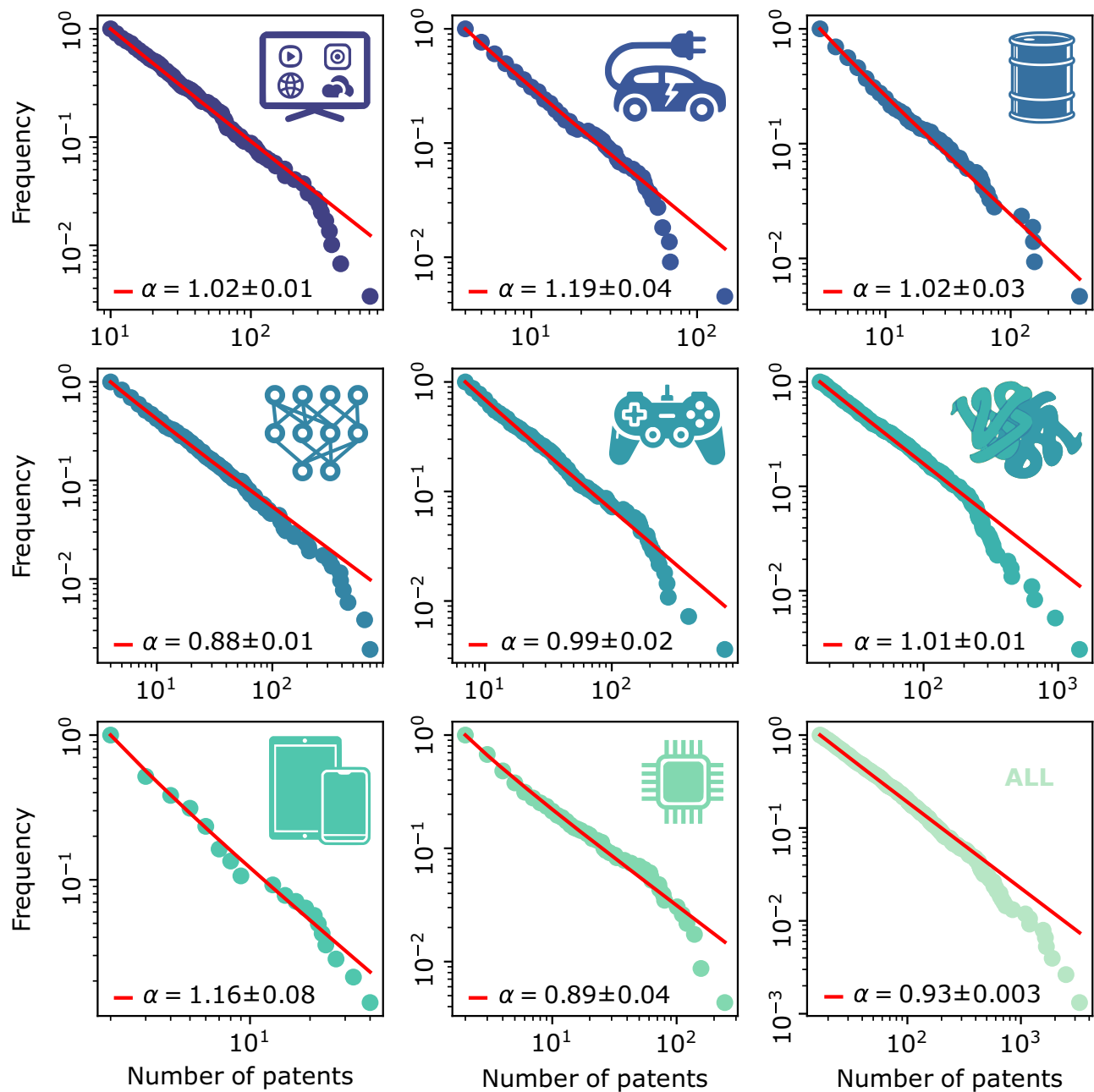


Fig. 1 | Zipfian distribution of patent output across U.S. locations and technological domains. Log-log plots showing the number of patents per location, ranked by total patent count, for eight geo-referenced datasets covering distinct innovation sectors: end-user applications, plug-in cars, fracking, machine learning, video games, biotechnology (enzymes), mobile devices, chip architecture, and the

combined dataset (top to bottom, left to right). In all cases, patent output follows a Zipfian scaling relationship with a power-law decay (2) and exponent γ close to 1. This pattern is consistent across domains, indicating a heavy-tailed distribution in which a small number of locations account for most patenting activity, while a long tail of locations contributes lower but non-zero output.

systems, altering the visible geometry of the innovation landscape. While the clustering hierarchy remains scale-free, the apparent inequality becomes scale-dependent. These findings show that Zipf's law, though robust in capturing size heterogeneity, is sensitive to the scale of observation—linking spatial inequality to broader urban scaling dynamics⁶.

Modeling the innovation landscape

To investigate the generative mechanisms behind the empirical trends in U.S. patenting, we develop a minimal stochastic model (Methods) in which innovation activity emerges from two general rules: local reinforcement and non-local dispersal. Despite its simplicity, the model can explain the characteristics of the innovation landscape without requiring sector-specific parameters or historical calibration.

Innovation unfolds through a spatial stochastic process that determines where new patents appear over time according to two mechanisms (Fig. 6a). The first mechanism, local reinforcement, increases the probability that a new patent will be assigned to a location with prior innovation activity. This behavior reflects a cumulative advantage mechanism similar to preferential attachment (9), leading to the establishment of dominant hubs and the observed rank-patent distribution (Zipf exponent ≈ 1), evidenced in and replicated by the model in Fig. 6d.

The second mechanism introduces non-local dispersal, which allows new innovation activity to appear at varying distances from established hubs. We model this using a Lévy-flight-like process, where most patents appear near existing locations, but occasional long-range “jumps” seed new regions. This assumption is supported by observations of fractal clustering

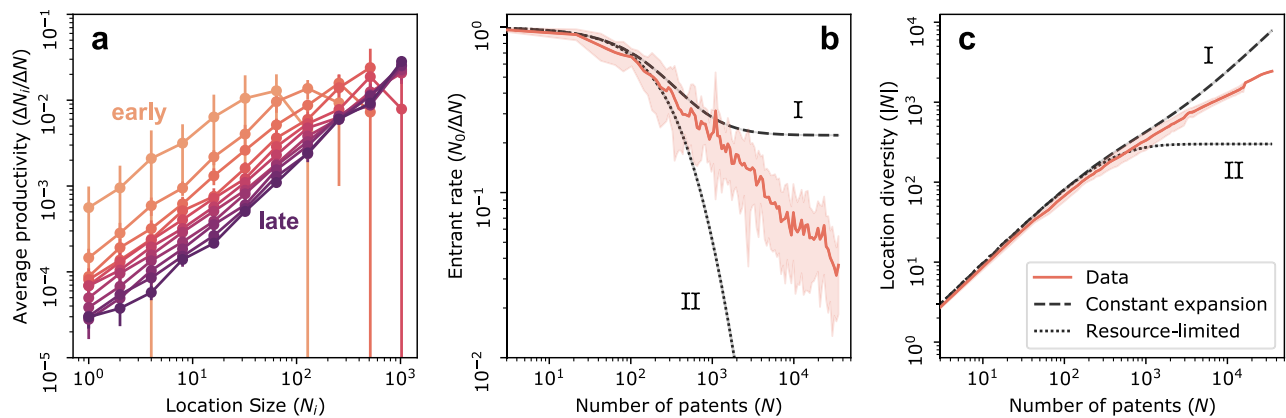


Fig. 2 | Cumulative advantage and entry dynamics in U.S. innovation.

a Relationship between prior patent count and subsequent innovation probability for biotech patents. Each point represents the average patenting rate for locations of size i , grouped by 10 equally sized temporal segments (deciles) of the patenting timeline. Color shades represent each decile, from early (light red) to late (dark red) periods. Error bars show standard deviations within each size category. The trend shows a size-dependent productivity pattern consistent with cumulative advantage, in which locations with more historical activity have

proportionally higher rates of future patenting. **b** The rate of new entrant locations—defined as those that produce a patent for the first time—measured across cumulative patent output for eight technological domains. The observed decline follows a power-law trend. Two model scenarios are included for comparison: constant entry rate (Scenario I) and resource-limited entry (Scenario II) (see text). **c** Growth in the number of unique patenting locations over time, compared to the model scenarios. Shaded regions in (b, c) represent standard deviations across domains.

in geometric networks, as well as dispersal mechanisms in ecological foraging, human mobility, and innovation networks^{19,34–36}.

The interplay of reinforcement and exploration generates synthetic innovation landscapes that qualitatively match empirical patterns (Fig. 6e vs. Fig. 3c, d). To systematically explore these dynamics, we examine how the balance between the two mechanisms—controlled by the model parameters p (reinforcement strength, see Algorithm 1) and v_0 (initial entry rate for new locations)—shapes patenting inequality (Fig. 6b). We find a continuous transition between distinct inequality regimes: high reinforcement and low dispersal produce concentrated landscapes dominated by a few innovation hubs (high γ exponent), while weaker reinforcement and more frequent long-distance entry lead to more distributed innovation activity and flatter rank-size distributions (low γ exponent).

Notably, the model predicts the emergence of a characteristic spatial scale at which inequality peaks before declining—a feature we refer to as the inequality horizon (Fig. 6h). This term denotes the aggregation radius R^* at which the coefficient of variation in patenting activity reaches its maximum. It captures a structural transition in the spatial organization of innovation, arising from the trade-off between local accumulation and regional diffusion—an emergent property of the model rather than an imposed constraint.

To support the model’s prediction of an inequality horizon, we derive an analytical condition for the aggregation scale R^* at which spatial inequality peaks (Methods). The expression (13) captures how the coefficient of variation, $CV(R)$, evolves with the aggregation radius R , drawing on both empirical patterns and the mechanisms revealed by our simulations. A simplified equation (15) shows that the inequality horizon emerges at the scale R^* where cluster growth and inter-cluster heterogeneity are in dynamic balance:

$$\mu_R^* = \mu_0 \left(\frac{\delta}{1 - \delta} \right). \quad (1)$$

Here, μ_0 is the threshold beyond which heterogeneity is amplified by spatial aggregation (see “Methods”), and δ is the Taylor exponent (12) quantifying how spatial fluctuations scale with mean cluster size³⁷. This closed-form prediction aligns closely with empirical data (see Fig. 6h), grounding the inequality horizon in a measurable trade-off between local accumulation and regional diffusion. As such, it provides a theoretical foundation for

interpreting the inequality horizon as an emergent, structural property of spatial innovation dynamics.

Discussion

Innovation is a complex phenomenon shaped by multiple interacting forces operating across spatial and temporal scales. While much of the literature emphasizes sector-specific conditions, institutional contexts, or policy interventions, it remains unclear what general organizing principles govern the geography of innovation. In this study, we adopted a complexity-science perspective to investigate whether innovation exhibits large-scale statistical regularities that transcend technologies and local policies.

Over a century of U.S. patent data reveals three robust patterns across technological domains: Zipfian scaling of innovation output, fractal clustering of innovation hubs, and a spatial scale—near 20 km—at which inequality in patenting peaks before declining. These results extend urban scaling theory by showing that innovation is not simply superlinear with city size, but distributed across multiple spatial scales. Innovation clusters nest within each other hierarchically, forming fractal patterns of inventive activity.

The persistence of Zipfian exponents and fractal organization across domains suggests that innovation growth is driven by the interaction of multiplicative, path-dependent dynamics and spatial diffusion. Yet while Zipf’s law is typically examined at the level of cities or metropolitan areas, this approach embeds assumptions about the spatial scale of innovation that may not align with its functional geography. By applying a multiscale spatial aggregation method, we reveal how inequality unfolds across space-independent of administrative boundaries—and identify a critical transition point. At scales below this threshold, cumulative advantage amplifies local disparities; beyond it, diffusion processes redistribute innovation more evenly. This scale-dependent peak in inequality defines what we call the inequality horizon—the spatial radius at which heterogeneity in innovation output reaches its maximum.

This interpretation is supported by a minimal stochastic model of innovation landscapes that combines local reinforcement with long-range, Lévy-flight-like dispersal. Without assuming sector-specific factors or institutional detail, the model replicates the empirical patterns. Notably, the inequality horizon is not externally imposed but emerges from the balance between concentration and diffusion. This provides a theoretical foundation for interpreting scale-dependent inequalities as measurable, robust features of innovation dynamics, arising from simple generative mechanisms.

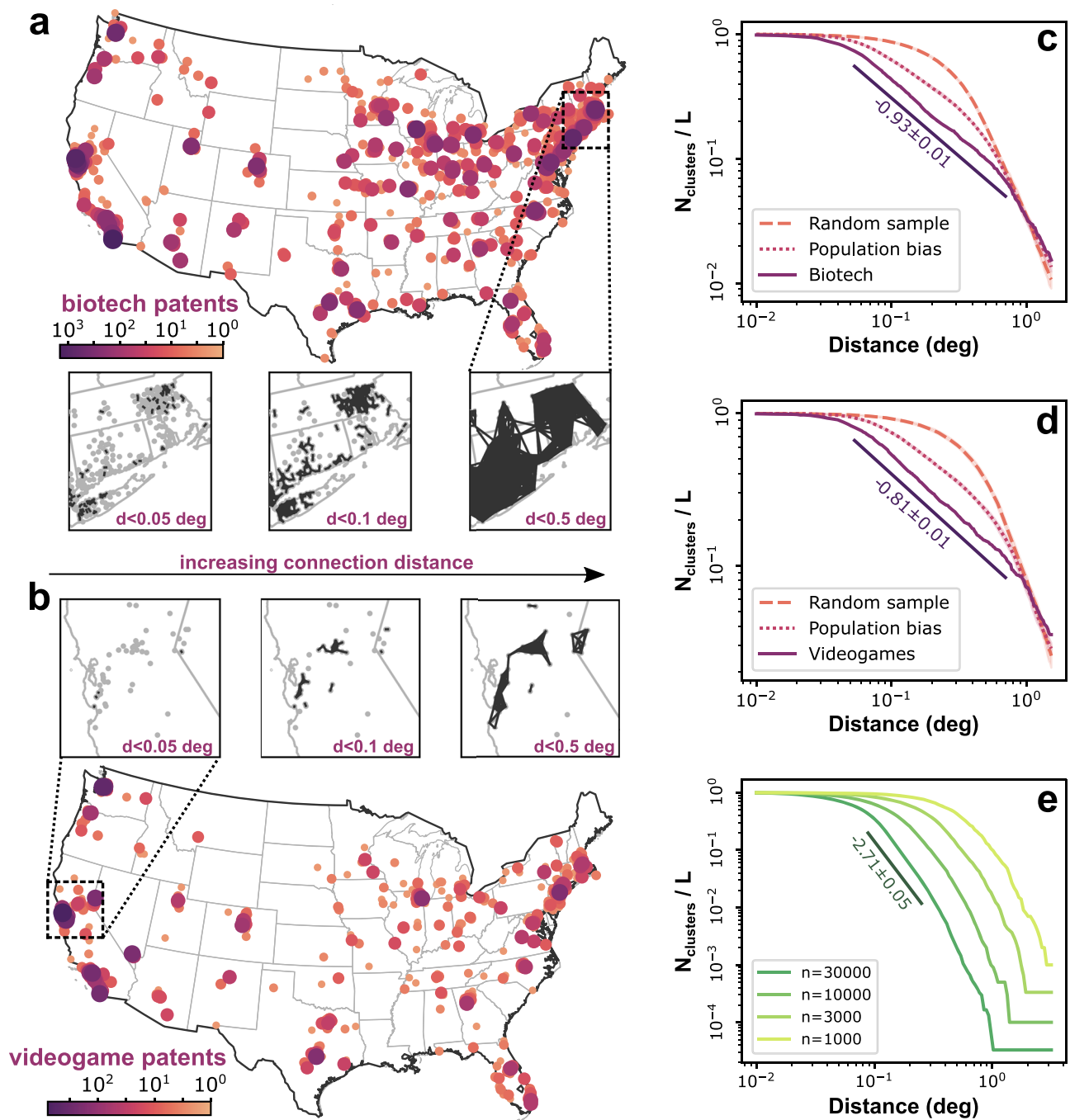


Fig. 3 | Multiscale structure of innovation networks in the United States. The spatial distribution of patenting activity is analyzed for two technological domains: **a** biotechnology and **b** video games. Insets show geometric graphs constructed from patenting locations, where nodes represent locations and links denote spatial proximity within a fixed radius (see text). The fraction of connected components (N_{clusters}/L) as a function of the aggregation radius (3), with power-law decay (4) in biotechnology (c) and video games (d) domains. Comparison with two null models

(see text): random down-sampling of U.S. cities (dashed line) and population-biased sampling based on demographic weights (dotted line). Shaded areas represent standard deviation across replicates. **e** The spatial distribution of innovative locations cannot be replicated using the US city distribution. The slope of the relationship (4) is unaffected by the various levels of downsampling of the entire city distribution.

Our study distinguishes between two forms of spatial inequality. Emergent inequality arises organically from self-reinforcing dynamics and spatial diffusion, as captured in our model. Directed inequality, by contrast, results from top-down interventions such as targeted investment, policy incentives, or the deliberate creation of innovation districts. Both forms may coexist in real-world systems, but our results suggest that even in the absence

of centralized planning, spatial disparities in innovation can arise spontaneously.

The peak in patenting inequality at approximately 20 km suggests that commuting distances play a central role in shaping innovation landscapes. This scale-regional rather than strictly urban-extends beyond municipal boundaries yet remains structurally significant at the national level.

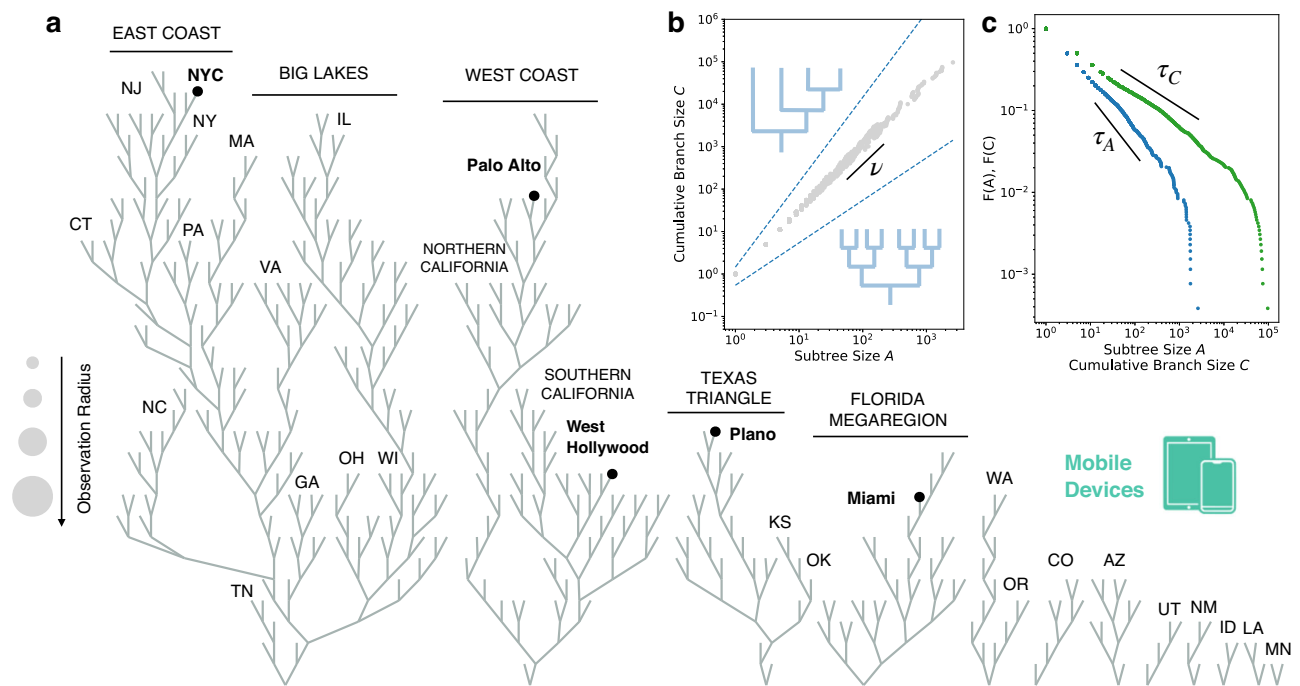


Fig. 4 | Hierarchical clustering and topological scaling of U.S. innovation networks. **a** Clustering tree of mobile device patents, showing the spatial merging of clusters across increasing aggregation radii, from 1 km (top) to 350 km (bottom). The subtrees merge into a single connected component beyond the percolation radius of $R_c \approx 2000$ km. These clusters align with U.S. mega-regions, demonstrating that localized innovation ecosystems remain structurally resilient across spatial scales. **b** Topological balance of the clustering tree, showing the relationship (5) between branch size (A) and cumulative branch size (C). A power-law relationship

(6) is observed in the empirical data (gray circles), with exponent $\nu = 1.44$ (black line). Reference curves show logarithmic (balanced) and quadratic (unbalanced) scaling (Methods). **c** The cumulative complementary distribution functions (CCDFs) for branch size $F(A)$ (blue) and total branch size $F(C)$ (green) both follow a power-law pattern (7), with fitted exponents $\tau_A = 1.6$ (blue circles, black fit line) and $\tau_C = 1.4$ (green circles, black fit line). All technological domains exhibit similar scaling behaviors.

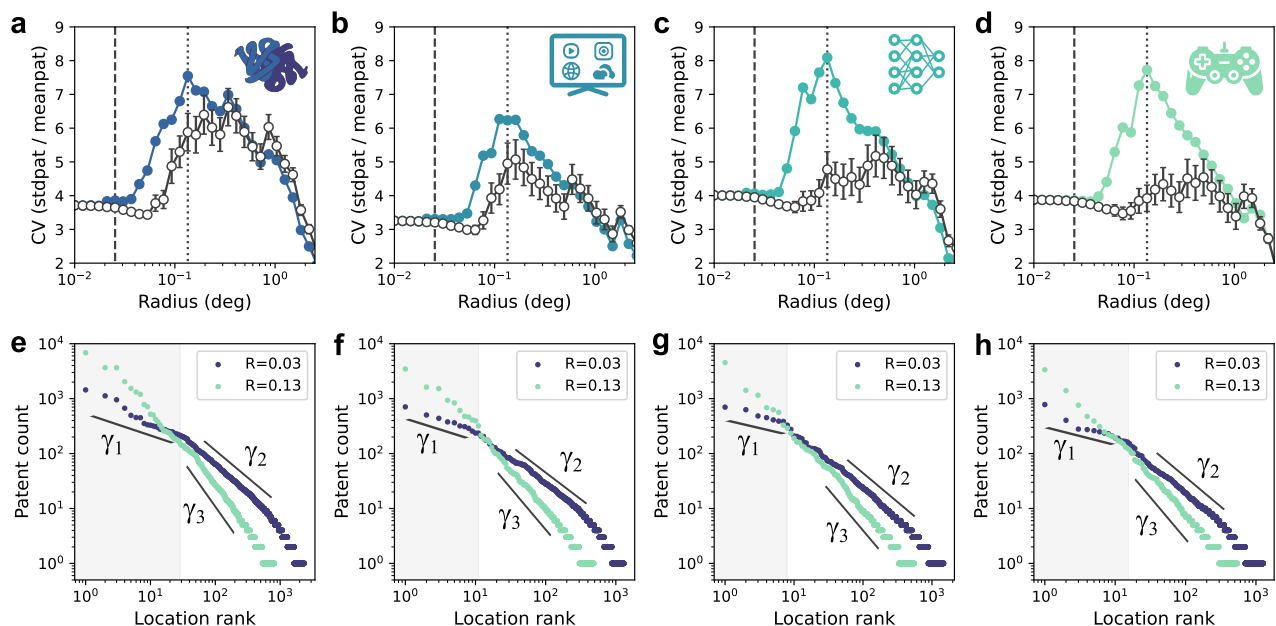


Fig. 5 | Scale-dependent spatial inequality and aggregation effects in U.S. patenting. The coefficient of variation (CV) in patent output across locations is shown as a function of aggregation radius R for the four largest technological domains: biotechnology (**a**), mobile applications (**b**), machine learning (**c**), and video games (**d**). CV is computed by aggregating patent counts within clusters of increasing radius R centered on each location and comparing standard deviation to the mean. Solid lines denote empirical values; open dots show the average of 100 randomized controls that preserve location coordinates but shuffle patent

counts. In all domains, CV increases with R , reaches a maximum near $R \approx 0.18^\circ$ (~ 20 km), and then declines toward the randomized baseline. **e–h** Zipf rank-size distributions of aggregated patent counts for the same domains, shown at $R = R_{\min} = 0.03^\circ$ (~ 3.3 km) and $R = 0.18^\circ$ (~ 20 km). At small scales, distributions exhibit a two-regime structure; at intermediate scales, they converge to a single power-law with steeper exponent $\gamma' > \gamma$, reflecting reduced heterogeneity due to spatial aggregation.

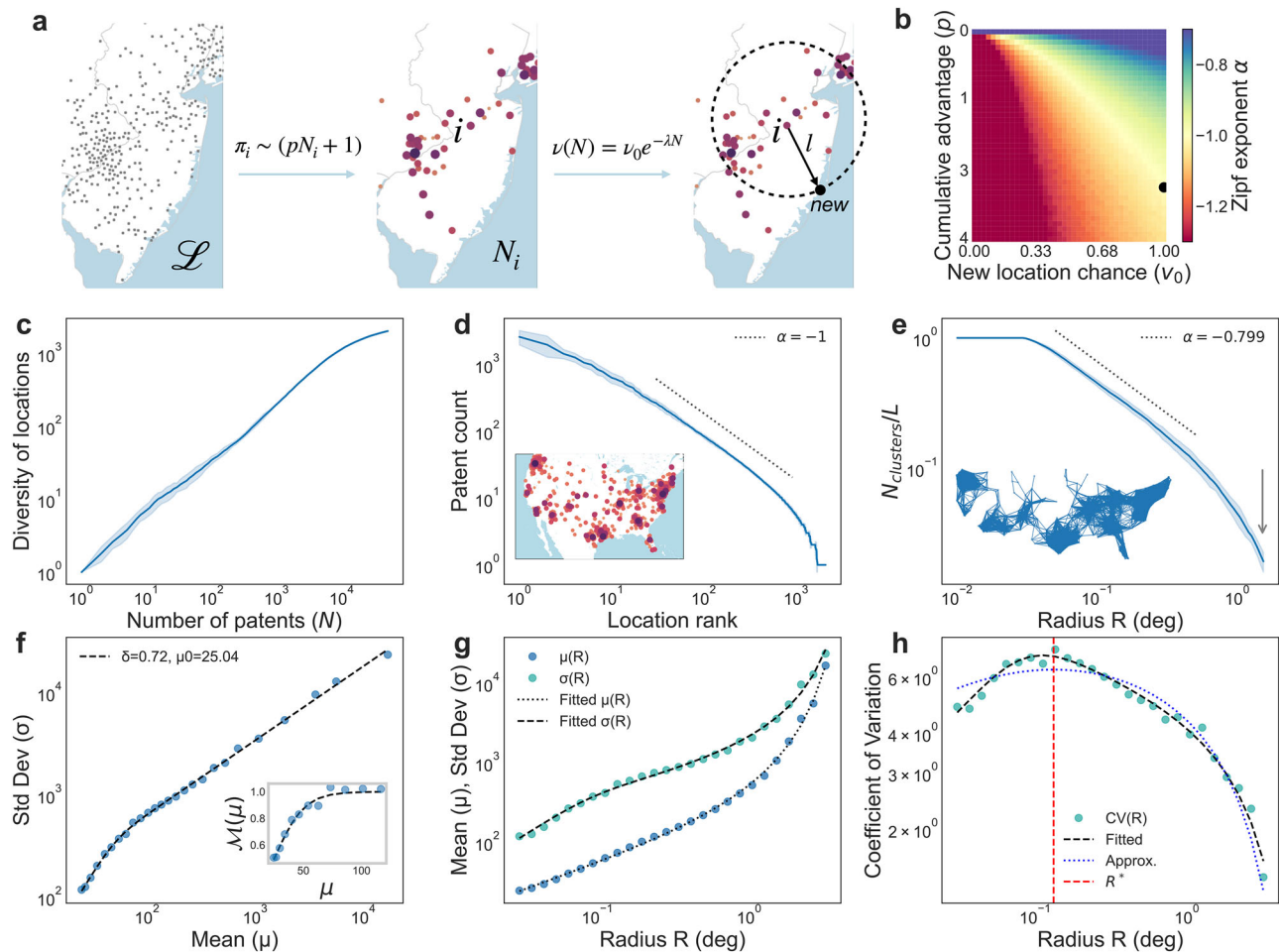


Fig. 6 | A stochastic model of innovation dynamics. **a** Each step assigns a new patent to a source $i \in \mathcal{L}$ selected with a probability π_i proportional to its productivity N_i . With probability $\nu(N)$, a new location is placed into the “urn” $\mathcal{L}$ via a Lévy “jump” of length l from i (see text). **b** The phase diagram illustrates the link between Zipf exponent, the probability of entering a new site (ν_0), and cumulative advantage strength (p). A transition from hub-centric inequality to more egalitarian distributions occurs as ν_0 increases or p decreases. **c** Simulated expansion of active locations based on total patent production N , aligning with actual entrant growth (Fig. 2b). **d** Simulated rank-size distribution of patenting across locations follows a power law with exponent $\gamma \approx -1$, as seen empirically (Fig. 1). The inset depicts the simulated distribution of patent counts. **e** The fraction of connected components (N_{clusters}/L) as a function of aggregation radius (R) follows a power

law (4) with exponent $\beta \approx -0.799$, agreeing with the empirical curves in Fig. 3c–e. The inset illustrates how most continental locations of the simulated graph G_R at $R \approx R_c$ belong to a single-connected component (vertical arrow). **f** The composite Taylor’s law (13) best fits the simulation data, predicting an exponent $\delta \approx 3/4$. Inset shows the empirical fitting of the logistic modulation function $\mathcal{M}(\mu)$ given by Eq. (14). **g** Analytical prediction of the growth of the average μ_R and standard deviation σ_R of cluster size with aggregation radius R (see text). **h** Multiscale dependency of the coefficient of variation $CV(R)$ with aggregation radius R in the simulation (filled circles). Dotted lines indicate two theoretical estimates of the CV curve and the predicted inequality horizon (vertical line), specified by Eqs. (13), (15), and (1) (see text). Simulation parameters (black dot in (b)): $p = 3$, $\nu_0 = 1$, $\lambda = 4 \times 10^{-4}$.

Commuting patterns influence labor mobility, supply chain networks, and knowledge spillovers, all of which affect where innovation occurs. These dynamics may have deep historical roots, linked to the spatial organization of cities since antiquity³⁸.

From a policy perspective, these findings imply that strategies focused solely on dense, centralized clusters may overlook the broader spatial structure of innovation. Supporting regional interconnectivity at scales around the inequality horizon—rather than concentrating exclusively on major urban cores—could help reduce disparities and strengthen innovation capacity. The multiscale approach introduced here provides a scalable framework for identifying where such leverage points lie, potentially informing infrastructure design, institutional support, and sub-city-level interventions.

At the same time, our approach has limitations. Patent data, while rich and geographically detailed, captures only certain types of innovation and omits informal, non-patented, or tacit knowledge. The analysis is limited to the United States, and innovation geographies elsewhere may follow different spatial patterns. Our model is deliberately minimal: it abstracts from sectoral complexity, institutional variation, and network structure to isolate

endogenous mechanisms. Yet despite this simplicity, it successfully reproduces the key empirical patterns through the interplay of cumulative advantage and spatial dispersal. Rather than offering a complete mechanistic account, the model demonstrates how complex spatial structures can emerge from domain-independent principles.

Framing innovation as a complex system may enable a more predictive and generalizable theory of technological change. Other complex phenomena—such as animal foraging, epidemic spread, and urban mobility—exhibit similar scale-dependent transitions where local reinforcement interacts with long-range connectivity. Just as scaling laws and generative mechanisms have revealed universal principles in physics, biology, and social systems, they may also help uncover fundamental regularities in the evolution of innovation across time and space.

Methods

Patent database

A comprehensive database was developed to analyze the spatial and temporal distribution of patent activity within a subset of contemporary U.S.

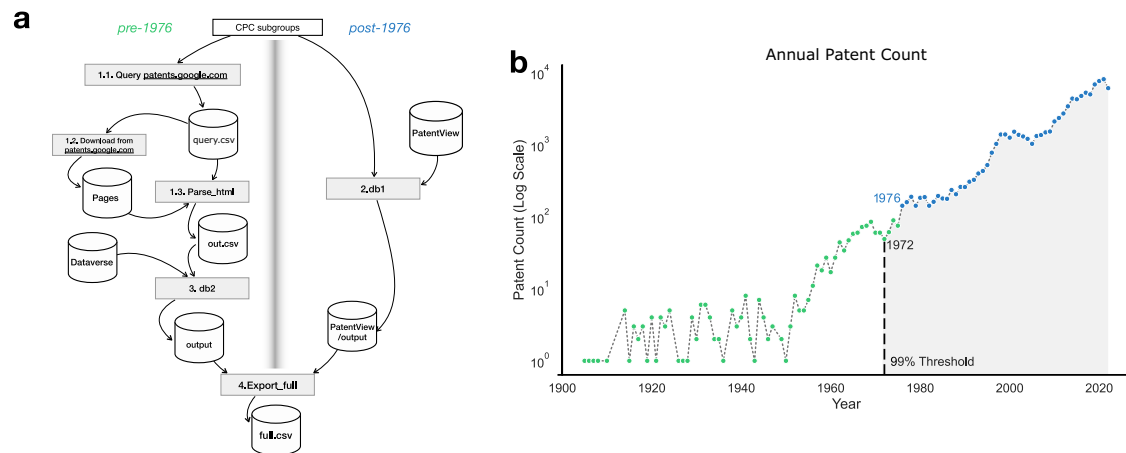


Fig. 7 | Construction of the patent database. **a** This diagram summarizes the multi-stage reconstruction process used to build a spatially and sectorally resolved dataset of U.S. patents. The pipeline processes more than 4 million patents and extracts approximately 140,000 records across 8 technological domains. Two distinct workflows are implemented: one for pre-1976 patents (left side), which rely on scanned image records and OCR-based extraction, and another for post-1976 patents (right side), based on structured digital records. Both workflows include sector-specific keyword filtering, geolocation of assignees and inventors, and

deduplication procedures (see “Methods”). **b** Annual counts of granted patents in the United States (1905–2024) are plotted on a logarithmic scale to emphasize long-term growth dynamics and visualize year-to-year variation across multiple orders of magnitude. The red dashed line indicates the year 1972, marking the threshold after which 99% of the total cumulative patents in the dataset were issued. The plot highlights the sharp rise in patenting activity over recent decades, which accounts for the majority of the dataset and motivates the focus on post-1970 dynamics in several analyses.

patents spanning from 1905 to 2024. We identified eight high-impact innovation sectors, i.e., plug-in cars, fracking, end-user multimedia applications, video games, chip architectures, machine learning, biotechnology enzymes, and mobile devices—that play pivotal roles in driving technological progress and economic growth within the United States. The United States Patent and Trademark Office (USPTO) maintains a high-quality data source called PatentView³⁹, which we have used to compile the database. Older patents (prior to 1976) were retrieved by manually downloading records from the USPTO’s public database (June 2024). This effort produced a unified, georeferenced patent dataset capturing contemporary technological trends across key innovative U.S. sectors, providing a valuable basis for both quantitative and spatial analyses of patent productivity over the past six decades. The resultant patent database comprises 139,387 USPTO patents.

For each patent, we collected detailed information, including the title, brief description, country of origin, assignee, names of inventors⁴⁰, and geographic coordinates (latitude and longitude) of the first inventor’s location. To categorize patents by technology type, we first identified Cooperative Patent Classification (CPC) codes that correspond to each sector. CPC codes serve as a standardized method of classifying inventions based on their technical characteristics and applications. We assigned patents to technology sectors by searching the entire patent database for these CPC codes, which represent high-value innovation sectors. Each patent was categorized into a sector if it contained at least one relevant CPC code in its documentation. In the following, we describe each technological sector, its CPC code criteria, and its relevance in the U.S. innovation landscape.

The End-User Applications (24,165 patents with CPC code H04N 21/47) sector comprises patents for end-user multimedia applications characterized by their interactivity, service structure, and diverse functionalities. Relevant patents cover a wide range of services, from local multimedia applications to high-bandwidth uplink technologies, showcasing the growing influence of digital media in consumer technology.

The Plug-In Electric Cars (9,320 patents with CPC code: Y02T 90/14) sector encompasses patents related to plug-in electric vehicle technologies, which contribute to the development of sustainable transportation. These patents reflect innovation within the electric vehicle industry.

The Fracking (4,609 patents with CPC code: E21B 43/16) sector covers patents for enhanced hydrocarbon recovery methods, specifically hydraulic

fracturing, used to mobilize hydrocarbons in reservoirs. Innovations in fracking technologies have transformed U.S. energy production and reshaped the nation’s role in global energy markets.

The Machine Learning (21,462 patents with CPC code: G06N 20/00) sector includes patents related to machine learning technologies, which equip machines with adaptive capabilities based on experiential data. These patents range from algorithms to physical machine implementations and reflect the transformative impact of artificial intelligence on diverse industries.

The Video Games (20,490 patents with CPC code: A63F 13/00) sector comprises patents encompassing a wide range of video game technologies, including hardware accessories, software innovations such as virtual camera animation, network-based game features, and cutting-edge game content generation, shaping the landscape of the video game industry and driving technological advancements. The dataset captures the technological complexity of modern electronic entertainment.

The Biotech Enzymes (51,853 patents with CPC code: C12Q) sector includes patents for processes involving enzymes, nucleic acids, and microorganisms, with applications in diagnostics and bioengineering. These patents reflect advances in healthcare, genomics, and environmental biotechnology, representing areas of life sciences innovation.

The Mobile Devices (2,108 patents with CPC code: H04M 1/0202) sector captures patents for portable telephones, including mobile phones and cordless handsets, specifically focusing on structural features of mobile devices. The growth of this sector illustrates the evolution of mobile communication and connectivity technologies.

The Chip Architecture (5,380 patents with CPC code: G06T 1/20) sector includes patents for graphics processors, including GPUs, pipelines, and architectures for image data processing. Patents in this sector are central to advancements in computer graphics, parallel processing, and the increasing performance demands of visual data applications.

Database reconstruction pipeline

The study recovers the patent database using a customized pipeline that combines many data sources and methods to deliver a complete georeferenced patent dataset. This reconstruction was designed to enable robust data collection, systematic processing, and integration of patent metadata (see Fig. 7a). First, by consolidating data from multiple sources, it ensures comprehensive coverage and minimizes biases inherent in individual

datasets. Second, separating records based on temporal and classification differences allows for better handling of data heterogeneity, such as the pre-1976 and post-1976 patent formats. Lastly, the inclusion of georeferenced inventor locations supports the spatial multiscale analyses that are central to this study.

The process begins with data collection from three main platforms: (1) Patent View, (2) Dataverse, and (3) patents.google.com. Each of these sources contributes distinct datasets that complement one another. For instance, Google Patents provides raw patent records divided into pre-1976 and post-1976 periods to account for differences in archival formats. Patent View adds enriched metadata fields, while Dataverse contributes supplementary datasets such as classifications and regional identifiers. Semi-automated Python scripts were used to interact with these platforms, querying and downloading records based on patent identifiers.

Once the data is retrieved, parsing routines extract relevant information from the HTML files obtained from patents.google.com. This structured data includes fields such as patent titles, inventor locations, and classification codes, which are consolidated into an intermediate database labeled db1.

In the next stage, the database is refined by integrating additional fields from PatentView and Dataverse, such as CPC subgroup classifications (see above), geospatial coordinates, and temporal data. This step enhances data granularity and completeness, resulting in an updated database, db2, that contains both the parsed information and the enriched metadata.

Finally, the fully processed dataset is exported into a standardized CSV file, labeled full.csv, for use in subsequent analyses. This final export contains all fields necessary for the study, ensuring compatibility with statistical and geospatial analysis tools. The CSV format facilitates its use in a wide range of applications, from hierarchical clustering and spatio-temporal pattern analysis to inequality measurements.

Our analysis spans the time interval from 1905 to 2024; however, the bulk of the patent publications are concentrated in the last 52 years. As shown in Fig. 7b, the cumulative number of patents surpasses 99% of the total dataset starting from the year 1972.

Fitting Zipf's law

To investigate the heterogeneous distribution of patent activity across U.S. innovation sectors, we use Zipf's law, a principle commonly used to analyze ranked distributions. Zipf's law describes a situation where the frequency $f(r)$ of an item is proportional to its rank r , expressed as

$$f(r) \sim r^{-\gamma}, \quad (2)$$

where γ is the exponent characterizing the distribution. By plotting patent counts against rank on a log-log scale, we can assess whether patenting activity follows Zipf's law across different geographic locations. For each sector, we ranked locations by patent count and fitted a power-law model using maximum likelihood estimation from the Python Powerlaw package version 1.5¹¹ (based on the work of Clauset et al.⁴²), to evaluate if the observed rank-frequency distribution adheres to Zipf's law or whether it can be best described by other distributions.

Analysis of size-conditional productivity

Temporal analysis of patenting activity reveals a two-phase pattern in the geographic diffusion of innovation. The initial phase is characterized by rapid spatial expansion, with each new patent frequently introducing a previously untapped location into the innovation network. This “boom” phase spans the emergence of the first 40–80 patents, regardless of technological domain (Fig. 2b), and is typically concentrated in established innovation hubs such as Silicon Valley (machine learning, software^{43,44}), Houston (fracking⁴⁵), and Boston (biotechnology⁴⁶).

As patenting activity increases, the number of new entrants gradually decreases. Secondary hubs, such as Austin (electric vehicles⁴⁷, gaming⁴⁸) and Raleigh (biotechnology^{49,50}), gradually enter the network, but do not displace

the dominance of primary centers. The consistent temporal pattern shows that general principles, in combination with sector-specific dynamics^{51,52}, shape the geography of innovation over time.

To quantify temporal dynamics, we analyze the size-conditional productivity of locations and the rate at which new locations enter the innovation network. We divide the patenting history into discrete time intervals, each containing 1/10th of the total patenting sequence, and measure patenting rates across these periods. Using a sliding window approach, we define a “present” bin containing 300 patents while considering all preceding patenting activity as the “past.”

For each location of size i (i.e., locations that had already produced i patents before the bin), we compute the normalized patenting productivity as:

$$P_i = \frac{\Delta N_i}{\Delta N}$$

where ΔN_i represents the increase in patent counts for locations of size i , and ΔN is the total number of patents in the interval. By removing P_0 data (productivity associated with locations that had never produced a patent before), we characterize the cumulative advantage effect (Fig. 2a). Fitting a power-law relation to the empirical productivity values yields an estimated exponent $\gamma = 0.962 \pm 0.021$, supporting the hypothesis that preferential attachment governs the growth of established innovation hubs.

In contrast, the entrant rate is quantified by measuring the fraction of patents associated with first-time entrants. Using sliding bins of 300 patents (with 25-patent step sizes), we track entrant rates over time across all eight technological sectors, obtaining an average trajectory.

To better understand the mechanisms driving these empirical trends, we consider two limiting cases in a stochastic innovation model: (1) the Scenario I (Constant Innovation Rate) assumes a fixed rate of new locations entering the innovation landscape. Under this scenario, diversity would grow linearly with the total number of patents, implying that patenting activity remains widely distributed without strong hub entrenchment; (2) the Scenario II (Resource-Limited Expansion) assumes that the system is bounded by an initial diversity pool, meaning that as patenting activity grows, new entrant opportunities diminish exponentially due to resource constraints and hub dominance.

To formalize these dynamics, we implement a stochastic model that simulates the evolution of the innovation landscape. The system begins with a pool of locations that have yet to produce patents. At each iteration, a location is selected based on cumulative advantage dynamics, meaning that locations with a strong history of innovation are more likely to continue patenting. When a patent is assigned, the location either already exists in the innovation network (reinforcing its status as a hub) or is a completely new entrant. Depending on the scenario, new locations are either introduced at a fixed rate (Scenario I) or drawn from a finite, resource-limited pool (Scenario II). Neither of these idealized cases fully captures the empirical patterns, as seen in Fig. 2b. The observed entrant rate declines sublinearly, following a power-law slowdown. This suggests that while innovation hubs consolidate their dominance over time, geographic expansion continues at a diminishing rate rather than abruptly ceasing⁵³.

To examine how entrant dynamics influence the overall diversity of the innovation landscape, we track the temporal evolution of location diversity. If the system were purely resource-limited, diversity would saturate due to rapid entrant decay. Conversely, a constant-rate model would predict a steady increase in diversity. By tracking entrant rates and diversity growth under these conditions, we observe two contrasting behaviors. First, if new locations continue to emerge at a constant rate, diversity grows linearly with total patent output, and innovation remains widely distributed rather than being concentrated in hubs. And second, if location entry is resource-limited, diversity plateaus over time, and patenting activity becomes highly concentrated in a small number of dominant hubs.

Neither of these idealized cases fully captures empirical patterns, which instead exhibit a power-law slowdown in entrant rates and a long-term, sublinear increase in diversity (Fig. 2c). This suggests that while hub entrenchment is non-negligible, geographic expansion persists over long time scales, albeit at a diminishing rate.

Multiscale graph analysis

We develop a multiscale spatial framework that integrates geometric graph theory, percolation analysis, and fractal scaling to characterize the spatial organization of patenting activity. This approach quantifies how connected components of innovation emerge and how their structural properties vary with spatial resolution. Given a set of geolocated patents $\mathcal{P} = \{(x_i, y_i)\}_{i=1}^N$, we deduplicate it to obtain L unique coordinates $\mathcal{L} = \{(x_j, y_j)\}_{j=1}^L$, each with an associated weight N_j denoting the number of patents at that location. For each spatial scale R , we construct a geometric graph $G_R = (V, E_R)$ where nodes represent unique locations and undirected edges connect nodes within distance R :

$$E_R = \{(i, j) \in V \times V \mid \| (x_i - x_j) \|_2 \leq R\}.$$

We compute the set of connected components $Q_1, Q_2, \dots, Q_{n_R}$ of G_R , where each Q_i contains all nodes linked through paths of edges not exceeding distance R . We define the “fragmentation function” as the fraction of connected components:

$$f(R) = \frac{N_{\text{clusters}}}{L} \quad (3)$$

where N_{clusters} is the number of connected components and $L = |V|$ is the total number of nodes. This function tracks how the geometric graph fragmentation decreases as spatial proximity increases.

Empirical data exhibit a power-law decay in the fragmentation function:

$$f(R) \sim R^{-\beta}, \quad (4)$$

indicating a scale-invariant, fractal-like structure in the spatial distribution of patenting³⁶. This behavior contrasts with conventional spatial null models, where $f(R)$ typically decays exponentially or exhibits a sharp percolation transition^{54,55}. In the fractal regime—where the above scaling law holds—a single dominant connected component does not emerge, and patenting remains distributed across a hierarchy of spatial components. Although fractal patterns have been documented in urban geography³⁰, the scaling observed in patenting networks emerges independently of population distribution, as demonstrated by the failure of population-based null models to reproduce the empirical (β) exponent (Fig. 3c–e).

To capture the full history of spatial aggregation, we track the merging of connected components across a range of radius thresholds and encode this information in a clustering tree. Initially, each node forms a singleton component. To efficiently identify spatial neighbors at each scale, we insert the points in $\mathcal{L}$ into a quadtree spatial index⁵⁶. As R increases, overlapping connected components are merged, and these events are recorded as branches in a hierarchical tree. Each node in the clustering tree corresponds to a connected component, and merging events generate parent nodes from their child components. This structure preserves the complete multiscale aggregation history, enabling structural analyses of spatial organization. Figure 4a illustrates an example clustering tree whose hierarchical architecture is consistent across spatial scales. To assess the topological self-similarity of clustering trees, we adopt the shape tree analysis proposed by Herrada et al.³². For each region i , we define a subtree S_i as the union of the connected component rooted at i and all its descendant regions. The size A_i is the total number of patents within S_i , and the cumulative

branch size C_i is:

$$C_i = \sum_{j \in S_i} A_j. \quad (5)$$

This measure reflects the extent and asymmetry of spatial aggregation. Balanced trees minimize C_i , whereas more asymmetric or chain-like trees yield larger C_i values. Herrada et al. demonstrated that for many empirical trees, C_i and A_i satisfy an allometric scaling law:

$$C \sim A^\tau, \quad (6)$$

with a universal exponent $\tau \approx 1.44$. We observe a similar exponent in the spatial organization of patenting (Fig. 4b), suggesting that the clustering trees obey similar scaling constraints to those found in biological or information networks. To further assess scale invariance, we examine the complementary cumulative distribution functions (CCDFs) of A and C :

$$F(A) \sim A^{1-\tau_A}, \quad F(C) \sim C^{1-\tau_C}, \quad (7)$$

both functions exhibit power-law tails (Fig. 4c), which supports the fractal nature of empirical clustering trees.

Coefficient of variation across scales

To explore whether spatial inequality in patenting exhibits scale dependence, we measure the coefficient of variation $CV(R)$ —the ratio of standard deviation to mean of patent counts—across different aggregation scales R . Using the same geometric graph framework described above, we compute the connected components $Q_1, Q_2, \dots, Q_{n_R}$ at each spatial scale. The total patent count of component Q_i is:

$$W_i = \sum_{v_j \in Q_i} N_j.$$

From the set $\{W_i\}$, we compute:

$$\mu_R = \frac{1}{n_R} \sum_{i=1}^{n_R} W_i,$$

$$\sigma_R = \sqrt{\frac{1}{n_R} \sum_{i=1}^{n_R} (W_i - \mu_R)^2},$$

and the coefficient of variation:

$$CV(R) = \frac{\sigma_R}{\mu_R}. \quad (8)$$

To assess robustness, we perform bootstrapping with 100 resampled datasets. The resulting confidence intervals (yellow curves in Fig. 5, top row) confirm that the observed peak in inequality is not an artifact of sampling variability, but rather a persistent feature of the spatial distribution of innovation. Our Python implementation automatically calculates the multiscale coefficient of variation $CV(R)$ from a set of geographical locations.

A stochastic model of innovation dynamics

To investigate the emergence of spatial patterns in U.S. patenting, we develop a minimal stochastic model that reproduces key empirical regularities described in Results. The model simulates the evolution of innovation activity through two fundamental processes: cumulative advantage and non-local dispersal.

The simulation unfolds over a geographic domain corresponding to the continental United States. It is initialized with the locations of the 80

most populous U.S. cities (population > 500,000), which act as potential starting points for innovation. At each time step, a new patent is added to the system. With some probability, the patent is assigned to an existing location i drawn from the current set $\mathcal{L}$ of active locations. The selection probability is given by a reinforcement mechanism:

$$\pi_i = \frac{pN_i + 1}{\sum_j (pN_j + 1)}, \quad (9)$$

where N_i is the cumulative number of patents at location i , and p controls the strength of cumulative advantage. This rule ensures that locations with a history of patenting activity are more likely to continue growing, reinforcing existing hubs over time¹².

With probability $v(N)$, a new location is introduced instead. The rate of new location entry decays exponentially as the system matures:

$$v(N) = v_0 e^{-\lambda N}, \quad (10)$$

where N is the total number of patents added so far, v_0 is the initial rate of entry, and λ controls the rate of decay. This reflects the empirically observed

decline in new geographic entrants as existing hubs accumulate more patents (see Fig. 2c).

When a new location is added, its spatial coordinates are determined by a Lévy flight originating from the most recent location. Lévy flights combine frequent short steps with occasional long jumps and are widely used to model exploratory dynamics in spatially embedded systems^{57,58}. Distances are sampled from a Lévy distribution (via `scipy.stats.levy`, with scale $s=0.1$), and angles are drawn uniformly from $[0, 2\pi)$. If the resulting location falls outside U.S. borders (defined by C_o), the jump is repeated until a valid point is found using Natural Earth 110 m landmass shapefiles⁵⁹.

The simulation proceeds for L iterations, producing a synthetic history of patenting locations and timestamps. This output is evaluated using the spatial methods introduced above, including rank distributions, fractal clustering, and the coefficient of variation $CV(R)$. By systematically varying parameters p and v_0 , we map a continuous space of innovation regimes, from highly concentrated to widely dispersed systems (see Fig. 6b).

This model can be interpreted as a spatial urn process with reinforcement and exploration²⁵. The pseudocode below summarizes the model's implementation and variable definitions.

Algorithm 1. Stochastic model of patenting with reinforcement and Lévy dispersal

input :

- p : reinforcement strength (cumulative advantage),
- v_0 : initial entry rate for new locations,
- λ : decay rate of entry probability,
- s : Lévy flight scale parameter,
- C_o : polygon representing valid geographic boundaries (U.S. mainland).

output : Simulated patenting history H of length L with spatial coordinates.

Initialize set of locations $\mathcal{L}$ using 80 largest U.S. cities (each with $N_i = 0$);

while length of $H < L$ **do**

Compute selection probabilities for each $i \in \mathcal{L}$ using: $\pi_i = \frac{pN_i + 1}{\sum_j (pN_j + 1)}$;

Sample location i from $\mathcal{L}$ according to π_i ;

Append location i to patenting history H and increment N_i ;

Compute $v(N) = v_0 e^{-\lambda N}$, where $N = |H|$;

if random number < $v(N)$ **then**

Get coordinates (x, y) of location i ;

Draw Lévy-distributed jump from (x, y) with scale s to get (x', y') ;

while $(x', y') \notin C_o$ **do**

Redraw Lévy jump from (x, y) until $(x', y') \in C_o$;

end

Add new location (x', y') to $\mathcal{L}$ with $N_{new} = 0$;

end

end

Analytical estimation of the inequality horizon

This section presents a macroscopic model for estimating the peak of spatial inequality in U.S. patenting activity—termed the inequality horizon (see above). The inequality horizon corresponds to the aggregation scale R^* at which the coefficient of variation (CV) across patenting locations reaches its maximum.

The coefficient of variation, denoted $CV(R)$, is defined as the ratio of the standard deviation σ_R to the mean μ_R of aggregated patent counts within clusters defined by a spatial radius R around each location. Modeling the behavior of μ_R and σ_R as a function of R is thus essential for understanding spatial inequality.

In the framework of Lévy geometric graphs (LGG)³⁶, μ_R corresponds to the expected size of the connected component a node belongs to, given a connection radius R . For small R , most nodes remain isolated, and μ_R follows a power-law scaling:

$$\mu_R \sim R^a,$$

where a is a local aggregation exponent that quantifies the early-stage growth of clusters. Higher values of a indicate faster local connectivity, typically arising from dense short-range interactions. The value of a is influenced by the Lévy exponent s in the step-length distribution $P(l) \sim l^{-s}$, where smaller s implies more spatial dispersion and suppressed local aggregation.

Unlike standard LGGs, the innovation dynamics modeled here include a reinforcement mechanism through node revisitation. At each time step, with probability $1 - v$, the process returns to an existing node (10), selected preferentially by accumulated patenting activity (9). This accelerates the growth of clusters relative to memoryless models, increasing the observed exponent a .

As R approaches a critical scale R_c , a percolation-like transition occurs, and μ_R diverges as:

$$\mu_R \sim |R_c - R|^{-b},$$

where b is a divergence exponent characterizing the sharpness of the connectivity transition. Due to reinforcement, R_c tends to be lower than in standard LGGs, and the divergence can be steeper—potentially resembling explosive percolation⁶⁰.

To describe the full range of $R < R_c$, we propose the following unified scaling ansatz:

$$\mu_R = c R^a |R_c - R|^{-b}, \quad (11)$$

where c is a scale-dependent prefactor reflecting the overall density of innovation nodes. This expression captures both the early growth regime and the divergence near criticality. Figure 6g shows a least-squares fit of Eq. (11) to the simulation data.

Empirical analysis reveals that the standard deviation σ_R scales with the mean cluster size μ_R following a power-law relation:

$$\sigma_R \sim \mu_R^\delta, \quad (12)$$

where δ is the Taylor exponent, a measure of spatial heterogeneity. This form is equivalent to the classical Taylor's law stated in terms of variance: $\text{Var} \sim \mu^{2\delta}$. We adopt the standard deviation form for clarity and analytic convenience. In this formulation, $\delta = 1/2$ corresponds to Poisson fluctuations; $\delta > 1/2$ indicates clustering; and $\delta < 1/2$ implies regularity or ordering. In our dataset, we consistently find $\delta \approx 3/4$, suggesting nontrivial correlations.

While Eq. (12) holds asymptotically, empirical deviations occur at low μ_R due to sampling noise and finite-size effects. Figure 6g shows that σ_R initially grows slowly before transitioning to a power-law regime. To

account for this behavior, we adopt a composite form:

$$\sigma_R \sim \mu_R^\delta \mathcal{M}(\mu_R), \quad (13)$$

where

$$\mathcal{M}(\mu_R) = \frac{1}{1 + \exp[-k(\mu_R - \mu_0)]} \quad (14)$$

is a logistic modulation term that suppresses variability for small cluster sizes (see inset in Fig. 6f). Here, μ_0 sets the inflection point of the transition, and k controls its steepness.

This analytical framework applies for aggregation scales $R > R_{\min} \approx 0.03^\circ$ (approximately 3.3 km), above which spatial heterogeneity becomes scale-dependent. Below this threshold, empirical data show that $CV(R)$ remains approximately constant (see Fig. 5), indicating a scale-invariant regime not captured by our model. This plateau suggests the presence of statistically homogeneous urban cores—consistent with longstanding urban design theories that define compact, walkable neighborhoods with diameters between 3 and 4 km³³. These stable substructures likely reflect deeply rooted spatial organization in historical city planning and may serve as foundational units in the broader innovation landscape.

To obtain an analytic prediction for the inequality horizon, we approximate the logistic term with a rational function:

$$\mathcal{M}(\mu_R) \sim \frac{\mu_R}{\mu_R + \mu_0}.$$

Substituting this into Eq. (13) and dividing by μ_R gives an explicit expression for the coefficient of variation:

$$CV(R) = \kappa \left(\frac{\mu_R^\delta}{\mu_R + \mu_0} \right), \quad (15)$$

where κ is a constant of proportionality.

To find the scale R^* that maximizes $CV(R)$, we treat $\mu_R = \mu$ as the independent variable and set the derivative to zero:

$$\frac{d CV}{d \mu} = \kappa \frac{d}{d \mu} \left(\frac{\mu^\delta}{\mu + \mu_0} \right) = 0.$$

Applying the quotient rule:

$$\frac{d CV}{d \mu} = \kappa \frac{\delta \mu^{\delta-1} (\mu + \mu_0) - \mu^\delta}{(\mu + \mu_0)^2}.$$

Setting the numerator to zero and dividing both sides by $\mu^{\delta-1}$:

$$\delta(\mu + \mu_0) - \mu = 0.$$

Solving for μ gives the critical mean cluster size:

$$\mu^* = \mu_0 \left(\frac{\delta}{1 - \delta} \right).$$

Substituting μ_R for clarity:

$$\mu_R^* = \mu_0 \left(\frac{\delta}{1 - \delta} \right). \quad (16)$$

This condition holds only if $\delta < 1$, ensuring a finite maximum. When $\delta \geq 1$, spatial heterogeneity grows faster than mean aggregation, and $CV(R)$ increases monotonically toward the percolation threshold.

Given the parametric form of μ_R in Eq. (11), we can numerically invert μ_R^* to find the corresponding inequality horizon R^* . The parameters a , b , c ,

R_∞ , δ , μ_0 , and κ are fit to empirical data using nonlinear regression. Our Python code implements this procedure and provides theoretical estimates of the inequality horizon for any spatially distributed dataset of innovation activity.

Data availability

The processed database used in this study is publicly available at Zenodo (<https://doi.org/10.5281/zenodo.17079686>) and mirrored at this website: <https://github.com/svalver/fractal-patenting>.

Code availability

The custom software tools developed in this study are archived at Zenodo (<https://doi.org/10.5281/zenodo.17079686>) and publicly available at the website github.com/svalver/fractal-patenting.

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Author contributions

Salva Duran-Nebreda (SDN) developed the agent-based model, executed the simulations, generated the figures, and performed data analysis. Blai Vidiella (BV) contributed to data analysis and visualization. Sergi Valverde (SV) designed the study, reconstructed the dataset, performed data analysis, developed the analytical approximation, and wrote the initial draft of the manuscript. R. Alexander Bentley (R.A.B.) reviewed the theoretical framework and edited the manuscript. All authors contributed to the review and editing of the final version.

Competing interests

The authors declare no competing interests.

Additional information

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